



FAKULTI SAINS KOMPUTER DAN TEKNOLOGI MAKLUMAT

PROJEK SARJANA MUDA
BORANG CADANGAN TAJUK PSM
PSM Title Proposal Form

Sesi/Semester (*Session/Semester*): **2526/1**

Arahan: Sila lengkapkan Bahagian A, B dan C kemudian serahkan borang ini kepada Panel Penilai pada masa pembentangan tajuk.

Instruction: Fill the form in Part A, B and C then give to the Evaluator Panel during title presentation session.

BAHAGIAN A: MAKLUMAT PELAJAR

PART A : STUDENT PROFILE

Nama (*Name*) : Ganesh Raaj A/L Arumugam

Tahun/Kursus (*Year/Course*) : **3** ~~BIS~~ / BIP / ~~BIW~~ / BIM / BIT

No. KP (*IC Num*) : 030115081283 *Passport Number* :
 No. Matrik (*Matric Num*) : DI230042 *Mobile Num* : 01127801583

E-mel (*Email Add*) : di230042@student.uthm.edu.my

BAHAGIAN B: KETERANGAN PROJEK

PART B : PROJECT DETAILS

Nama **Penyelia** : Dr. Suhaila binti Mohd. Yasin
 (*Supervisor Name*)

Tajuk Projek : Smart Laboratory Booking and Resource
 (*Project Title*)
 Management System for FKMP UTHM

Sila lampirkan latar belakang projek cadangan yang mengandungi perkara berikut:
 (*Please attach project details as follows:*)

1. **Pengenalan** (*Introduction*)
2. **Penyataan Masalah** (*Problem Statement*)
3. **Objektif** (*Objective*)
4. **Skop Kajian** (*Scope*)

Keperluan Projek (*Project Requirements*):

Perisian (*Software*) : Visual Studio Code, XAMPP

Perkakasan (*Hardware*) : Laptop

Technology/Technique/
Method/Algorithm : Centralized laboratory and asset management, payment tracking and reporting, and role-based access control.

BAHAGIAN C: AKUAN PELAJAR

PART C : STUDENT ACKNOWLEDGEMENT

Saya mengakui bahawa tajuk di atas adalah :

I confirm that this project is:

- [] Idea saya sendiri
My own idea
- [✓] Dicadangkan oleh penyelia saya Dr. Suhaila binti Mohd. Yasin
Proposed by the supervisor.....
(Please write down the supervisor name)

Tarikh (Date):17/10/2025

Tandatangan (Signature):Ganesh.....

BAHAGIAN D: AKUAN PENYELIA

PART D : SUPERVISOR ACKNOWLEDGEMENT

I Dr. Suhaila binti Mohd. Yasin.....confirm that

I have reviewed this student's project proposal and therefore agree for the proposal to be submitted for evaluation.

Date :

Signature :

Official Stamp

1.0 Project Background

Laboratory management refers to the systematic organization and oversight of laboratory resources, including equipment, materials, scheduling, and facility maintenance. It plays a crucial role in supporting teaching, experimentation, and research activities in academic institutions. In universities, especially within faculties such as Mechanical and Manufacturing Engineering, laboratories are essential for hands-on learning, testing, and innovation. These environments require effective coordination of user access, safety protocols, and equipment usage to maintain operational efficiency and uphold educational and research standards. Efficient laboratory management contributes directly to institutional productivity, user satisfaction, and academic excellence (Rihm, 2024; Sun & Gao, 2025).

At present, the Faculty of Mechanical and Manufacturing Engineering (FKMP) at Universiti Tun Hussein Onn Malaysia (UTHM) relies on manual or semi-digital systems for managing laboratory access, bookings, and equipment usage. Laboratory users—including students, researchers, and lecturers—typically request access by filling out paper forms or contacting lab personnel directly via email or in person. Records of equipment usage, asset inventories, and maintenance schedules are maintained using logbooks, spreadsheets, or isolated digital documents. This decentralized approach involves multiple personnel and lacks standardized procedures or shared platforms for consistent data access or coordination (Argento, Rula, & Cowan, 2020; Yin et al., 2022).

The reliance on manual and unintegrated systems has led to several challenges within FKMP laboratories. Scheduling conflicts, double bookings, and underutilization of equipment frequently occur due to the absence of a centralized booking system. Maintenance tracking is often overlooked, increasing the risk of equipment breakdowns and reducing overall operational uptime. Data is scattered across various documents and managed by different individuals, making retrieval slow and prone to errors. Additionally, the lack of automated notifications or digital logs creates inefficiencies in approving bookings and tracking asset conditions. These issues hinder effective decision-making, delay laboratory access, and increase the administrative workload of academic and technical staff (Yin et al., 2022; Rihm, 2024).

To address these challenges, a Smart Laboratory Booking and Resource Management System is proposed for FKMP at UTHM. The system will be a centralized, web-based platform that streamlines laboratory booking, asset tracking, and payment processing through an integrated interface. It will incorporate intelligent automation features such as real-time scheduling, automated reminders, and a virtual assistant powered by natural language

processing (NLP) to guide users. The solution will also support proactive maintenance through integrated reporting tools and IoT-based monitoring, allowing staff to track equipment usage and condition more effectively. This approach is expected to improve resource utilization, minimize administrative delays, and promote a more transparent and user-friendly laboratory environment (Sun & Gao, 2025; Rihm, 2024).

2.0 Problem Statement

The current laboratory management process implemented within the Faculty of Mechanical and Manufacturing Engineering (FKMP), Universiti Tun Hussein Onn Malaysia (UTHM), relies heavily on manual and semi-digital procedures for booking, resource allocation, and record-keeping. While functional at a basic level, this system presents several operational challenges that hinder efficiency, transparency, and resource optimization (Yin et al., 2022; Rao et al., 2024).

At present, the laboratory booking process is conducted primarily through manual methods such as physical forms, email requests, or direct communication with laboratory personnel. This fragmented approach often results in scheduling conflicts, overlapping bookings, and delays in approval. Laboratory staff must manually verify availability and record bookings, which increases administrative workload and the likelihood of human error. If this issue remains unresolved, it will continue to limit the efficient utilization of laboratory resources, reduce user satisfaction, and impede the faculty's ability to manage laboratory operations systematically.

The absence of a centralized digital platform for managing laboratory assets and equipment information has led to difficulties in tracking the status, location, and maintenance history of resources. Equipment details are often stored in individual spreadsheets or logbooks maintained separately by each laboratory, resulting in data redundancy and inconsistency. This lack of integration hampers decision-making regarding equipment allocation, maintenance scheduling, and resource planning (Parks et al., 2022; Rao et al., 2024). If not addressed, this problem could lead to asset mismanagement, underutilization of laboratory equipment, and an increased risk of equipment failure due to inadequate maintenance monitoring.

For projects that require material usage fees or external access, payments are typically handled manually, with limited documentation and verification mechanisms. This manual approach complicates financial tracking and makes it difficult to ensure transparency and accountability in transactions. Without a proper digital payment and tracking system, there is a risk of data loss, unauthorized usage, and discrepancies between laboratory usage and

recorded income. If this issue persists, it may undermine administrative integrity, reduce financial accountability, and negatively affect the faculty's ability to manage its laboratory operations efficiently.

If these issues are not addressed, FKMP's laboratory operations will continue to face inefficiencies, data fragmentation, and administrative burden. The lack of automation and integration will hinder effective coordination among users and staff, ultimately affecting the quality of teaching, research, and resource management within the faculty. Therefore, the development of a Smart Laboratory Booking and Resource Management System is crucial to ensure centralized data management, improve operational efficiency, and enhance transparency in laboratory utilization and payment processes (Rao et al., 2024; Yin et al., 2022; Yuen et al., 2025).

3.0 Objective

The objectives of this system are to:

1. Analyze and design a Smart Laboratory Booking and Resource Management System using an object-oriented approach.
2. Develop the Smart Laboratory Booking System using web-based approach with XAMPP
3. Test the developed system using alpha and beta testing

4.0 Scope

This project focuses on the development of a Smart Laboratory Booking and Resource Management System within the field of laboratory management and scheduling. The study is conducted at the Faculty of Mechanical and Manufacturing Engineering (FKMP), Universiti Tun Hussein Onn Malaysia (UTHM), where multiple teaching and research laboratories operate under diverse functions that necessitate systematic management of bookings, equipment utilization, and maintenance activities. Effective coordination of these processes is essential to ensure optimal use of laboratory facilities, minimize scheduling conflicts, and maintain the operational efficiency of available resources.

Information required for the system design is obtained through consultations with laboratory technicians, academic staff, and administrative personnel who are directly involved in laboratory coordination, asset maintenance, and resource allocation within FKMP. The users of the proposed system comprise several groups: administrators, responsible for managing user accounts, configuring system parameters, and overseeing overall laboratory operations;

laboratory technicians, tasked with equipment registration, maintenance scheduling, and approval of booking requests; academic staff, who reserve laboratory equipment for teaching and research purposes and monitor resource availability; and students or external users, who submit booking requests, process payments, and utilize laboratory facilities according to approved schedules.

The scope of the system is delineated through its core functional modules, as presented in **Table 4.1**. These modules are designed to facilitate key processes such as laboratory booking, asset management, payment handling, maintenance tracking, and reporting. Each user category is granted specific access privileges to the modules relevant to their respective roles and responsibilities, ensuring secure, efficient, and role-based interaction with the system.

Table 4.1: Smart Laboratory Booking System's Module

Module	Description	User
User Authentication and Access Control	Provides secure login and role-based access management for different types of users including administrators, technicians, staff, and students or external users.	Admin, Technician, Staff, Student/ External User
Laboratory and Asset Management	Allows administrators and technicians to register, update, and track laboratory equipment, including specifications, availability, and maintenance history.	Admin, Technician
Booking and Scheduling	Enables users to view equipment availability, submit booking requests, and manage time slots. Includes automated approval workflows and conflict prevention.	Staff, Student/ External User, Technician, Admin
Payment and Billing	Manages payment processes for both internal and external users, including automated invoice generation and payment tracking to ensure financial transparency.	Staff, Student/ External User, Admin

Maintenance and Service Management	Allows technicians to log maintenance requests, monitor repair progress, and record service activities to ensure equipment reliability and safety.	Technician, Admin
Reporting and Notification	Generates analytical reports on laboratory usage, payments, and maintenance activities while providing automated notifications for approvals, upcoming bookings, and system alerts.	Admin, Technician, Staff, Student/ External User